

September 2026

Dear Editor-in-Chief,

I am pleased to submit the manuscript “An Aharonov–Bohm Lattice Operator Framework for the Riemann ζ Zeros: A 38-Test Computational Verification Suite” for consideration in your journal as an interdisciplinary computational-physics contribution. The manuscript presents the complete verification record of the AB-Cloud v23 SUPERCOMBO suite — the 31,239-line Julia 1.12 apparatus of the AB-Cloud research laboratory — which subjects an Aharonov–Bohm lattice-operator framework for the Riemann ζ zeros to thirty-eight independently designed computational tests, each with pre-registered acceptance windows.

The reference run executed the suite end-to-end in a single session of 26.04 hours and produced 79 verdict entries, an audit tier of 35 deep-audit batteries totalling 127 individually logged sub-checks, 60 publication-grade figures, per-test computation logs, and a consolidated final report. The framework couples an exact Gram-point/ ζ -zero dictionary at $\alpha = 1/2$ with a proven $\pm E$ chiral pairing, an AB lattice Hamiltonian whose flux sector encodes $\delta_C = \pi/7$, and a family of closed-form constants — the spinorial phase $\arg \gamma^* = 89.874^\circ$, the fractal factor $|CF| = 1$, and the half-factorial normalization $(1/2)! = \sqrt{\pi}/2$ — that tie the spectral data to the Gaussian unitary ensemble fingerprint. The record is decisive where the framework claims exactness: hermiticity, flux quantization, chiral pairing, the zero-mode index, and the identity chain hold at machine-zero residuals, and the rational identities are verified at 256-bit precision through independent computational routes.

Three points may assist editorial assessment. First, the claims are explicit and falsifiable: the framework reproduces the GUE fingerprint of the zeta zeros at effect size — every class-discriminating statistic (nearest-neighbour ratios under four protocols, number variance, spectral rigidity, the binned gap-ratio law) places both the lattice spectrum and the ζ dictionary on the GUE side of its pre-registered window — and every strict null rejection is accompanied by an effect-size window that the same data satisfy, the standard reading for sample sizes of $n = 50,000$. Second, the manuscript's post-run estimator audit traces the run's one adverse composite statistic (pooled two-sample $D = 0.1096$) to a documented reference-unfolding artifact and corrects it in the v3.2 reference build, with the falsifiable post-fix window $D = 0.03\text{--}0.06$ registered in the text. Third, the verification apparatus itself is a contribution: the adversarial two-pass design, the audit-tier sub-check grammar, and the calibration gates are documented to be re-used on any candidate spectral construction.

Since the package was frozen, the pre-registered post-fix window has been resolved by an independent follow-up: the standalone laboratory re-run (20 realizations, 91,360 pooled spacings, canonical ζ -side unfolding) gives pooled $D = 0.0970$ — above the pre-registered falsification threshold of 0.08 — so the residual distance is promoted, by the manuscript's own decision rule, to a measured physical effect. The September 2026 addendum (new final chapter of the manuscript and both preprints, with the raw runs shipped in the package) documents the cross-language reproduction of every pooled statistic, a disorder sweep and a finite-size probe showing the effect is neither disorder-

nor size-induced, and an effective GUE+Poisson decomposition that accounts for the full distance by an intrinsic $\approx 25\text{--}28\%$ Poisson admixture, with the post-fit residual equal to the ζ -vs-GUE calibration floor (0.0241 vs 0.0242). The composite verdict GUE-CONSISTENT stands on its pre-registered terms.

The submission is a complete verification package. The manuscript (PDF and editable DOCX, with the LaTeX preprint source) reproduces every verdict line verbatim in Appendix A; the complete v3.2 reference source is reproduced in Appendix B and shipped as a standalone file with its full audit changelog; the complete reference-run archive — all 38 test folders, the four-format final report, the consolidated execution log, the one-line verdict register, and the 60 figures — ships inside the package at `run_data/run_20260914_234626/` with a reading guide (Appendix E of the manuscript); and a Python reimplementation executes the suite's statistical semantics on commodity hardware in minutes. All materials are openly available at <https://github.com/wild8highlander/ab-cloud-research> (research record: <https://github.com/wild8highlander/research-papers>), and the archived dataset carries DOI 10.5281/zenodo.21825394.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests and no funding sources. I confirm that all code and data are openly available under the repository links above, and that the ORCID identifier is 0009-0003-7299-0701.

Thank you for your consideration. I would be glad to respond to any editorial or reviewer questions.

Yours sincerely,

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